ESG-451 Weekly Status Update
PhytoPi

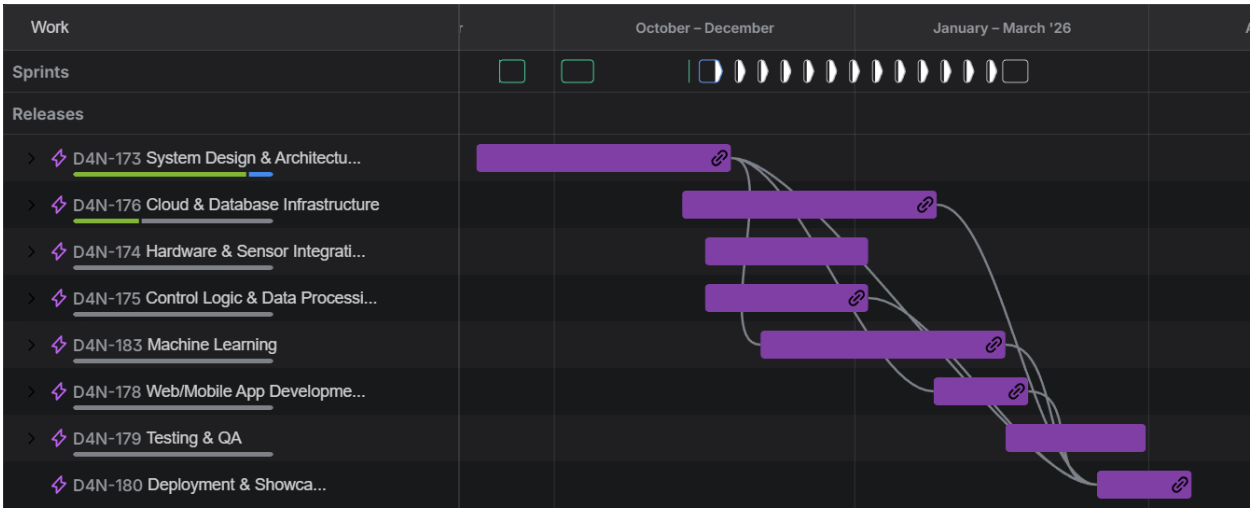
1. Project Description

PhytoPi is an IoT-based controlled environment system for supporting a plant throughout its entire growth cycle with minimal human intervention. Utilizing embedded hardware and a machine learning system monitoring the plant using a camera, PhytoPi is designed to automatically care for more than 80% of a plant's required maintenance over its lifespan.

2. Project Status Summary — Percent Complete 18%

The PhytoPi project is on-track for completion on the projected timeline. Currently, the Hardware/Integration Lead is streamlining the timeline for the Jira and improving the backlog/sprint layout and working on filling out order forms for ordering parts off the BOM. 3D models are being developed in Solidworks for 3D printing tests the next two weeks. The Software/Cloud Lead is working on frontend UI for the kiosk (screen available on the enclosure), and for the mobile app.

3. Timeline Snapshot



4. Current Concept

The current design concept featured in *Figure* 1 is an improvement over version 0.2 of the System Architecture by offering a basic logical flowchart of sensor data collection to cloud storage. In *Figure 2* we created a prototype of our physical product architecture. *Figure 3* features an electrical schematic and physical layout of devices wired to the main controller. *Figure 4* features the current UI design for the kiosk, web, and mobile devices, improving

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from version 0.1 by adding a dark theme and designs for each screen layout. *Figure 5* and *Figure 6* feature the database ERD and schemas respectively.

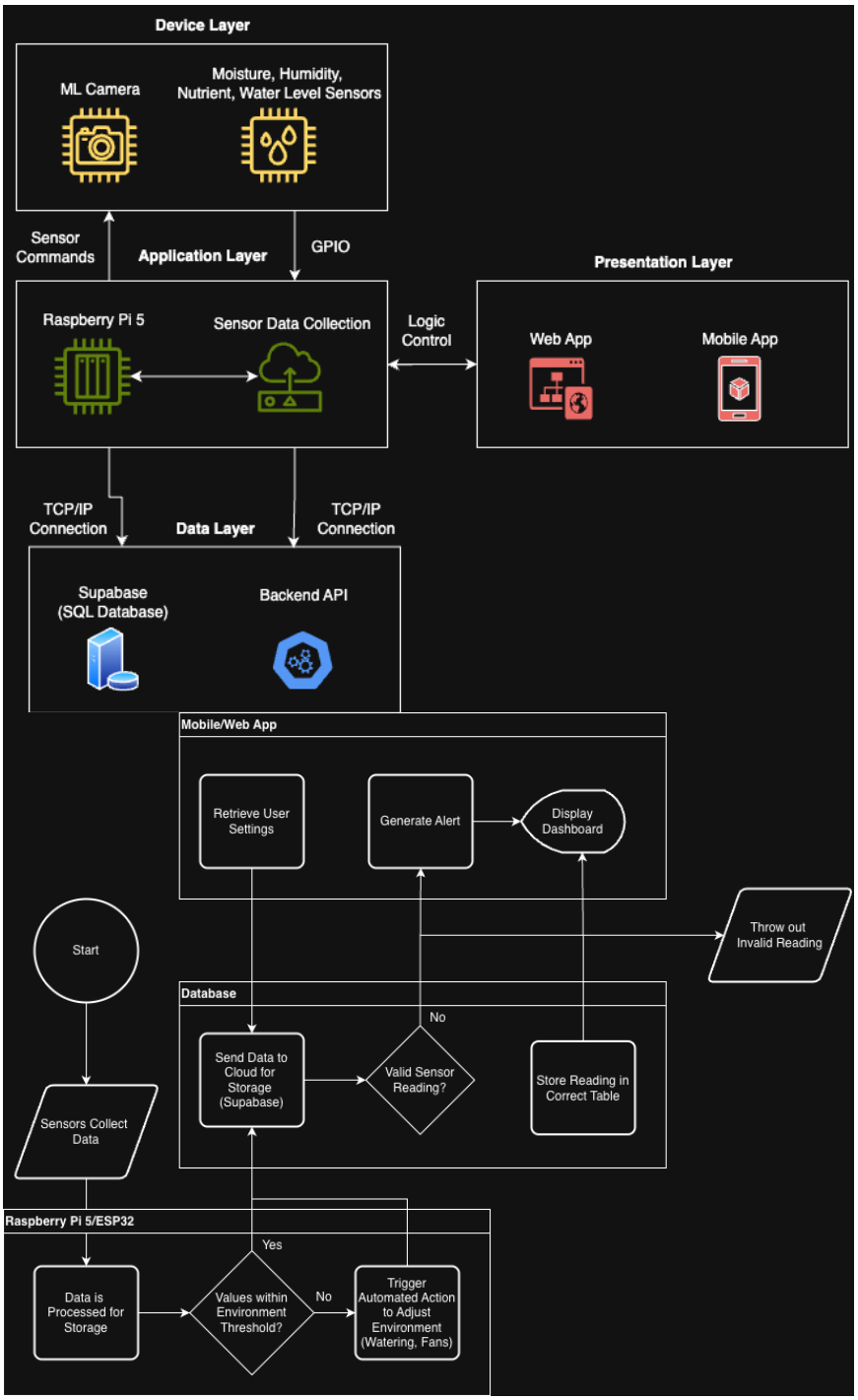

Figure 1. Current Design Concept - System Architecture Version 0.3

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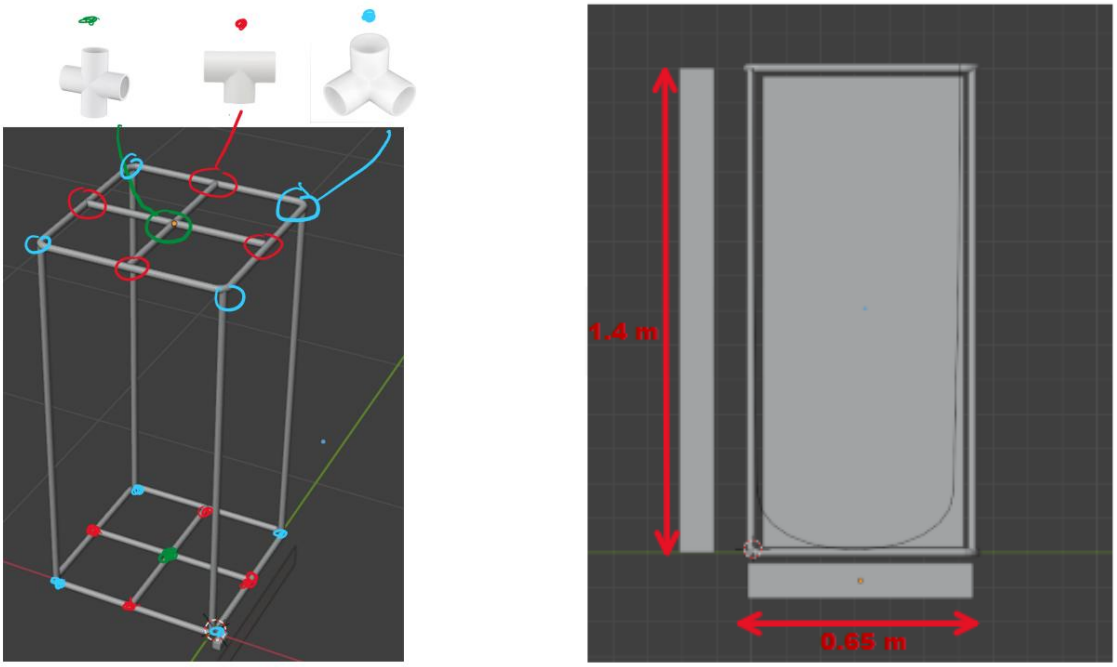

Figure 2. Current Design Concept – 3D Model of Final Product v0.1

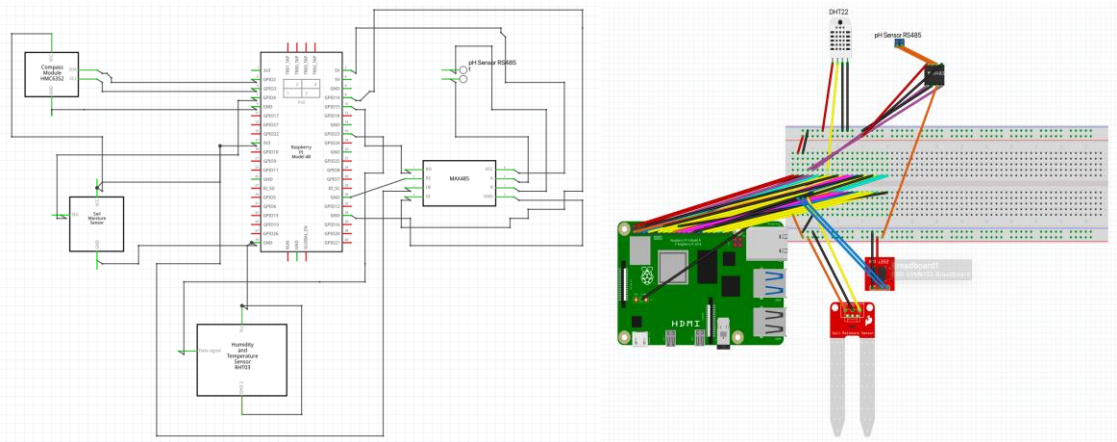

Figure 3. Current Design Concept – Wiring Diagram v0.3

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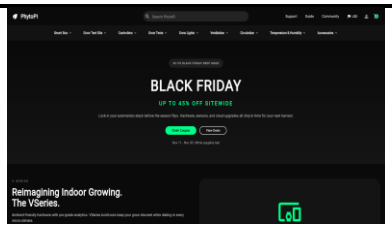
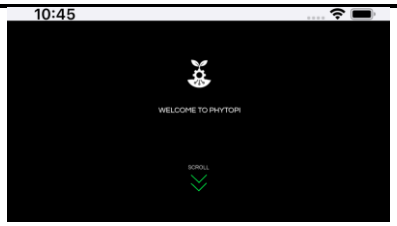
	Web	*Kiosk*	*Mobile*
Landing Page			

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Figure 4. Current Design Concept – UI Wireframe v0.2

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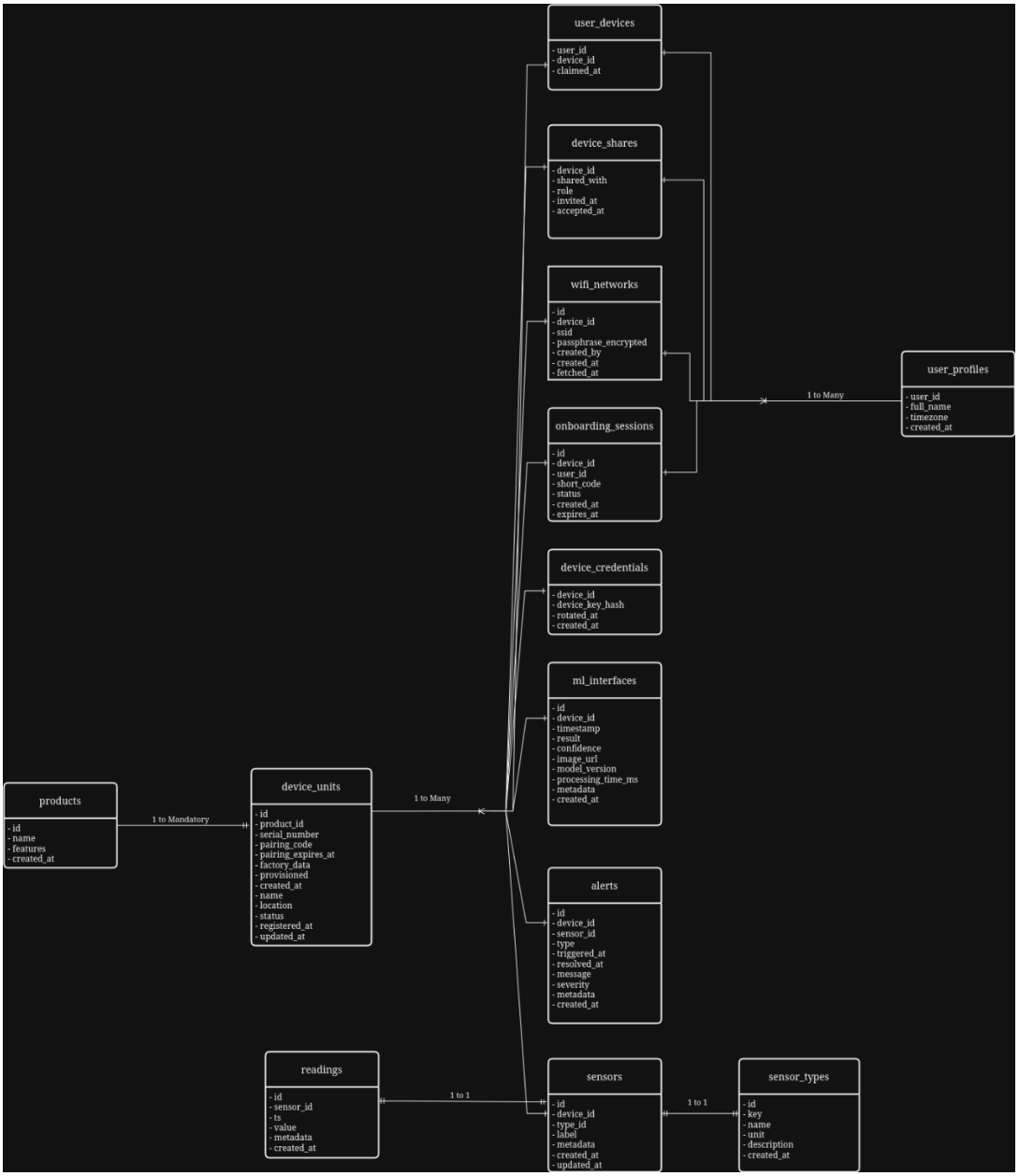

Figure 5. Current Design Concept - Database ERD v0.1

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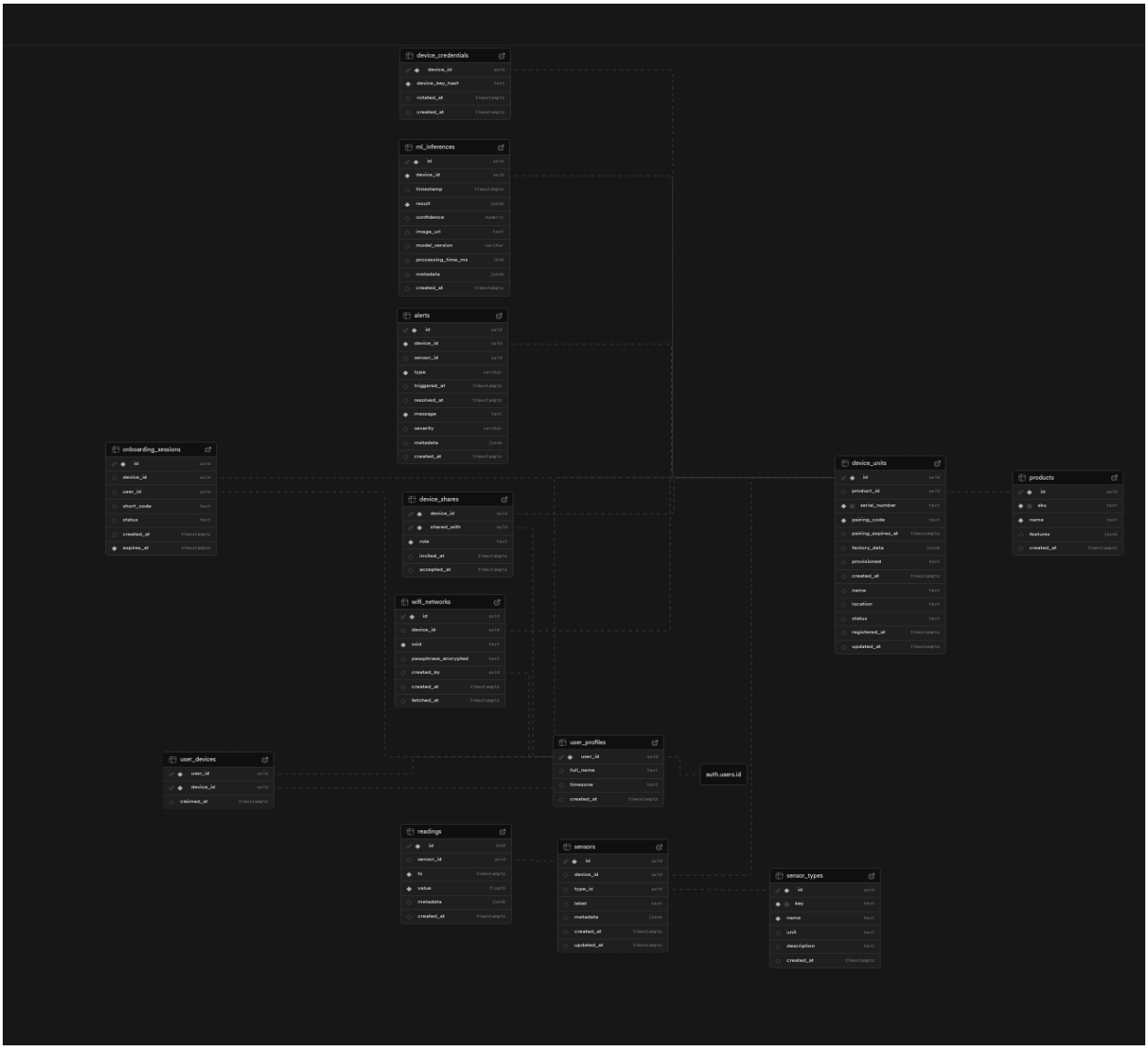

Figure 6. Current Design Concept – Database Schema v0.1

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5. Accomplishments and Learning

Planned from last week:

- UI Design – completed
- UI Development in Flutter – 15% completed
- Further Jira Grooming/Improvement – 90% completed
- Privacy agreement write-up – 10% completed
- Test Plan Documents – 5% completed
- Plant Pot Solidworks Design – 5 % completed
- Place BOM Order – In Progress

Unplanned:

- Soil Moisture Collection Logic
- Temperature/Humidity Collection Logic
- Light Intensity Collection Logic
- Soil Moisture Collection Logic
- Send Sample Readings to Database

Learning:

- Clear communication between team members is necessary for critical path items that require completion and distribute workload effectively.
- Document storage should be uncluttered and very clearly titled, with no duplicates or differing filetypes, and all documents should be kept locally as well as in cloud storage.

6. Work Planned for Next Week

- Critical Path – Solidworks modeling for the plant pot to prototype 3D-printed solutions for compatibility, place order for parts to build showcase.
- Finish test plan document according to IEEE standards
- Test and calibrate all available sensors for accurate data collection/compatibility.
- Set row-level rules for temperature and humidity values in database.
- Complete privacy agreement write-up.

7. Risks

- CAD design for plant pot might prove to be more difficult than expected with the lack of 3D modeling experience between both team members. This can be mitigated by making rapid prototyping for this a critical path for the project and emphasizing the importance of effort into this design.
- Without careful consideration of the entity relationships within the database, it may be the case that the database structure does not function as intended for data storage or does not follow the intended structure of our tables. This would require refactoring of the database structure, resulting in a time delay of up to one week.

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8. Deliverables and Milestones

Milestone	Planned	Forecasted	Actual	Status
Database Infrastructure	11/30/25	11/30/25	N/A	Complete
Complete System Design	10/31/25	11/1/25	N/A	Complete
Complete Initial Sensor Testing	11/30/25	11/30/25	N/A	On Schedule
Complete Solidworks Modeling	11/30/25	11/30/25	N/A	On Schedule
Deliverable	Planned	Forecasted	Actual	Status
Project Portfolio	10/31/25	10/31/25	10/31/25	Complete
Technical Specifications	11/2/25	11/2/25	11/3/25	Complete
Database Entity Relationship Diagram	11/1/25	11/1/25	11/2/25	Complete
Timestamped Sample Readings for Temperature	11/31/25	11/31/25	N/A	On Schedule

9. Attachments

N/A

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Approvals:

Team Role	Print Name	Signature	Date
Product Manager/Hardware + Integration Lead	Nolan Tuttle	Nolan Tuttle	11/17/25
Software + Cloud Lead	Daniel Grijalva	Daniel Grijalva	11/17/25
Faculty Mentor	Bill Hughes	Bill Hughes	11/17/25

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